Simon Bernier

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EDUCATION

McGill University, Faculty of sciences, Montreal, Canada

M.Sc. in Physics (Graduate Research Program) | **GPA: 3.94/4.00** September 2019 – August 2022
B.Sc. in Honors Physics, Minor in Mathematics | GPA: 3.88/4.00 September 2015 – April 2019

AI SAFETY & MACHINE LEARNING CONTEXT

Awards: Received a six-month (July to December 2026) Career Development and Transition Funding from Coefficient Giving.

Courses: Completed BlueDot's "Intro to Transformative AI" and "Technical AI Safety" course, focusing on AI alignment, governance, and technical safety challenges.
Participating in the September 2026 cohort of the Iliad Intensive, providing a broad overview of AI alignment research for mathematicians, theoretical physicists, and theoretical computer scientists.

Deep Learning: Rigorous self-study of *Deep Learning* (Goodfellow et al.), covering linear algebra, probability, neural network architectures and deep learning fundamentals.

Technical Projects: ARENA curriculum (self-taught). BlueDot's *Technical AI Safety Project.*

RESEARCH AND PROFESSIONAL EXPERIENCE

Collège Jean-de-Brébeuf, Montréal, Canada January 2024 – present
CÉGEP Édouard-Montpetit, Longueuil, Canada October 2023 – June 2024

CEGEP physics (college level)
Communicated complex physical principles to students, a key skill for AI safety outreach and documentation. Organized scientific conferences, served on environmental committees and participated in program reviews, demonstrating leadership and multidisciplinary collaboration.

McGill University, Montréal, Canada

Research Assistant – Quantum Materials May 2019 – January 2024
Simulated chaotic quantum systems and dynamic phase transitions using C++ and tensor methods, optimizing for high-performance computing environments. Implemented parallel algorithms to handle complex physical simulations. Processed and visualized research data for publication in peer-reviewed journals. Studied relativistic phase transitions in metamaterials in the terahertz domain through MATLAB modeling and in artificial quantum materials with the ITensor library.

Teaching Assistant May 2019 – December 2022
Lectured and led tutorials for undergraduate and graduate physics courses; managed grading and curriculum support for advanced topics in physics.

TECHNICAL SKILLS

Python (PyTorch), C, C++, Matlab, Julia, Mathematica, LaTeX, git/GitHub and Linux.

PUBLICATIONS

S. Bernier and K. Agarwal. "Spatiotemporal quenches for efficient critical ground state preparation in the two-dimensional transverse field Ising model**.**" *Physical Review B***, 111**, *054311*. **(2025)**

S. Bernier and K. Agarwal. "Spatiotemporal Quenches in Long-Range Hamiltonians. " *Physical Review B*, **108**, *024310*. **(2023)**

A. W. Schiff-Kearn, L. Gingras, **S. Bernier**, et al. "Front-induced transitions control THz waves." *Communications Physics*, **4,** *162* **(2021)**

S. Bernier, A. Huang, W. Reisner and A. Bhattacharya. "Evolution of Nested Folding States in Compression of a Strongly Confined Semiflexible Chain. " *Macromolecules*, **51**, *11*. **(2018)**

LEADERSHIP & VOLUNTEERING

Elite Athlete: Middle-distance runner (10-15 hours/week training). Team Captain for the McGill Redbirds (2019-2020), managing team logistics and coach-athlete relations.